DM42 PROBABILITY, STATISTICS & REGRESSION

From Counting and Chance to Measurement, Data Analysis and Prediction

A practical keystroke handbook

DM42 and Free42 editions • validated examples • reusable data and programs

How to use this handbook

This is a field guide, not a catalogue of keys. Each topic starts with the decision to make, gives the mathematics in compact form, walks through the exact RPN entry, and ends with a check. The worked data are supplied as CSV files, and the numerical answers are independently reproduced by `validation/validate.py`.

The notation is literal: f means the yellow shift key; STAT means the physical top-row key carrying the yellow STAT legend; × is the multiply key; and a boxed menu label means the corresponding softkey. Thus f × COMB opens PROB and selects combinations. XEQ appears only when you really execute a named program or function. In program mode, pressing XEQ records an `XEQ` instruction on the current program line.

Results here assume ordinary real mode and the default stack. Display rounding never changes the internal calculation. Set a useful display with f DISP; four to six decimals is usually enough for inspection.

Watch this

Menus are not subroutines. Do not prefix shifted keys or menu choices with XEQ. For example, use f × RAN, not XEQ RAN; use f STAT MEAN, not XEQ MEAN.

Result order matters

The current DM42/Free42implementation places the first statistic in X and the second in Y. MEAN returns $\bar{x}$ in X and $\bar{y}$ in Y; SDEV returns s_x in X and s_y in Y; SUM returns Σx in X and Σy in Y. If you only entered one variable, use the X result and ignore Y.

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Part I

Counting and Chance

CHAPTER 1

Counting without listing

Counting is the machinery under finite probability. First ask: *does order matter?* If yes, use permutations; if no, use combinations.

1.1 Factorials

For a positive integer, $n! = n(n-1)\cdots 2\cdot 1$, with $0! = 1$. The number of possible inspection orders for six instruments is $6! = 720$.

Keystrokes

6 f × N! $\rightarrow 720$

1.2 Permutations: order matters

The number of ordered selections of r items from n is

$$ {}_nP_r = \frac{n!}{(n-r)!}. $$

Assigning three distinct bench positions from eight technicians gives ${}_8P_3 = 336$.

Keystrokes

8 ENTER 3 f × PERM $\rightarrow 336$

The calculator consumes n from Y and r from X.

1.3 Combinations: order does not matter

The number of unordered selections is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

Selecting three samples from eight gives $\binom{8}{3} = 56$.

Keystrokes

8 ENTER 3 f × COMB $\rightarrow 56$

Reasonableness check

A permutation must be at least as large as its corresponding combination. Here $336 = 3! \times 56$.

CHAPTER 2

Probability rules that earn their keep

Probabilities lie between 0 and 1. For mutually exclusive outcomes, add. For independent stages, multiply. For "at least one," the complement is often shortest:

$$P(A^c) = 1 - P(A), \qquad P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If events are independent, $P(A \cap B) = P(A)P(B)$. Conditional probability is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

2.1 A reliability example

Three independent series components have reliabilities 0.98, 0.97 and 0.995. The system works only if all three work:

Keystrokes

.98 ENTER .97 × .995 × → 0.945847

Two independent parallel units, each with reliability 0.92, work unless both fail:

$$1 - (1 - 0.92)^2 = 0.9936.$$

Independence is an engineering assumption, not a button. A common power rail, environment or maintenance error can make failures dependent.

CHAPTER 3

Binomial decisions

A binomial model has a fixed number n of independent trials, two outcomes per trial, and constant success probability p. Then

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

3.1 Quality-control lot

Suppose each of 20 items is defective with probability 0.025. Let X be the number defective.

$$P(X = 0) = 0.975^{20} = 0.602688,$$

$$P(X = 1) = 20(0.025)(0.975)^{19} = 0.309071,$$

$$P(X \geq 2) = 1 - P(0) - P(1) = 0.088242.$$

Keystrokes

For $P(X = 1)$: 20 ENTER 1 f × COMB, then multiply by .025 and by .975 ENTER 19 y^x.

Reasonableness check

The mean defect count is $np = 0.5$, so zero and one should dominate. An 8.8% upper tail is plausible.

3.2 Communications availability

If a packet succeeds independently with $p = 0.98$, the probability of at least nine successes in ten is

$$10(0.98)^9(0.02) + (0.98)^{10} = 0.983822.$$

For an upper tail, add the few high-end terms; for a long tail, use the complement of the shorter side. Do not round p between terms.

CHAPTER 4

Sampling without replacement and rare events

4.1 Hypergeometric sampling

When a sample is drawn without replacement, trials are dependent. If a lot of N items contains K defectives and n are sampled,

$$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}.$$

For $N = 50$, $K = 4$, $n = 5$, the probability of finding at least one defective is

$$1 - \frac{\binom{46}{5}}{\binom{50}{5}} = 0.353040.$$

Evaluate each combination with f × COMB. Preserve the denominator on the stack or in a numbered register.

4.2 Poisson counts

For independent rare events occurring at mean rate λ,

$$P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}.$$

With $\lambda = 2.4$ alarms per hour,

$$P(X \geq 4) = 1 - \sum_{k=0}^{3} e^{-2.4}\frac{2.4^k}{k!} = 0.221277.$$

Use e^x for the exponential and f × N! for $k!$. A Poisson model is not appropriate if the rate changes sharply or events arrive in bursts.

CHAPTER 5

Random numbers and simulation

5.1 RAN and SEED

f × RAN returns a pseudorandom value U with $0 \le U < 1$. Enter a nonzero seed, then press f × SEED. This produces a repeatable sequence. Seed 0 requests a time/platform-derived seed; a negative seed is treated by magnitude.

To simulate a fair die, compute $\lfloor 6U \rfloor + 1$. The supplied program is:

Listing 5.1: DIE: one simulated die roll

```
LBL "DIE"
RAN
6
x
IP
1
+
RTN
```

Run it with XEQ, select `DIE`, then ENTER. The result is an integer from 1 through 6.

5.2 Monte Carlo π

Generate pairs (x, y) uniformly on the unit square and count those satisfying $x^2 + y^2 \le 1$. With m hits in N trials,

$$\hat{\pi} = 4m/N.$$

Typical error shrinks only like $1/\sqrt{N}$: 1,000 trials are roughly ten times more stable than 10. The method is excellent for intuition and awkward regions, but inefficient when an exact formula is available.

5.3 Redundancy by simulation

For a two-unit parallel system, generate two independent U values. A unit with reliability p works if $U < p$; the system fails only if both comparisons fail. Compare the simulation estimate against $1 - (1 - p)^2$. A large mismatch usually signals a comparison or counter error before it signals new physics.

Part II

Measurements and Descriptive Statistics

CHAPTER 6

Statistical registers and safe data entry

6.1 Choose the register set

By default, the calculator accumulates the core sums in R11–R16. The ALLΣ layout uses R11–R23 and adds sums such as $\Sigma x^2 y$ needed by advanced work; LINΣ returns to the compact layout.

Keystrokes

f STAT ↓ ALLΣ or LINΣ

Changing the layout changes which registers are occupied. Avoid storing unrelated values there.

6.2 Clear, add, remove

Before a new dataset:

Keystrokes

f ← CLΣ

For one variable, enter each value in X and press Σ+. To remove an exact mistaken observation, re-enter it and use f Σ+, which is Σ−.

For a pair (x, y), use the calculator's required order:

Keystrokes

y ENTER x Σ+

That leaves x in X and y in Y when the pair is accumulated. Reversing them silently swaps the regression axes.

6.3 Matrix entry

A two-column real matrix may be placed in X and accumulated with [Σ+]; column 1 is x, column 2 is y. This is convenient in Free42 and for data prepared elsewhere. Confirm column order with a two-row test before loading a large matrix.

CHAPTER 7

Mean, spread and summaries

For observations $x_1, \dots, x_n$,

$$\bar{x} = \frac{1}{n}\sum x_i, \qquad s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}.$$

The calculator's SDEV is the sample standard deviation ($n-1$ denominator), not the population standard deviation.

7.1 Resistor batch

Enter the 12 values in `data/resistors.csv` using [Σ+].

Keystrokes

Mean: [f] [STAT] [MEAN] $\rightarrow \bar{x} = 10.0050$ kΩ in X.
Spread: [f] [STAT] [SDEV] $\rightarrow s_x = 0.020671$ kΩ in X.
Sum: [f] [STAT] [SUM] $\rightarrow \Sigma x = 120.06$ kΩ in X.

The range is 9.97 to 10.04 kΩ. Mean and standard deviation summarize centre and spread; they do not prove normality or conformance.

7.2 Accuracy is not precision

Five readings from instrument A have mean 10.200 and $s = 0.01581$; five from B have mean 10.000 and $s = 0.11895$. Against a true value of 10.000, A is precise but biased; B is accurate on average but less precise. A single statistic cannot describe both properties.

CHAPTER 8

Weighted means and pooled reporting

When values x_i represent unequal counts or importance weights w_i,

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}.$$

Enter each pair with the weight in Y and value in X:

Keystrokes

w_i ENTER x_i $\Sigma+$, then f STAT WMEAN

For three production shifts with means 9.98, 10.02 and 10.01 from 8, 15 and 7 pieces, the weighted mean is

$$\frac{8(9.98) + 15(10.02) + 7(10.01)}{30} = 10.0070.$$

Do not average group means without weights unless the group sizes are equal.

CHAPTER 9

Outliers and defensible summaries

The 15 shaft diameters in `data/shafts.csv` give $\bar{x} = 20.003067$ mm and $s = 0.012092$ mm. Removing the final 20.041 mm value gives $\bar{x} = 20.000357$ mm and $s = 0.006234$ mm.

That contrast is diagnostic, not permission to delete. Investigate transcription, instrument state and process history. Report both analyses when exclusion is not justified. The sample mean changes little, but the standard deviation nearly doubles, which strongly affects capability and prediction.

Part III

Paired Data and Regression

CHAPTER 10

Correlation and the straight-line model

For a linear fit, the calculator models

$$\hat{y} = mx + b.$$

Correlation r measures the strength and direction of a linear association; it is unitless and lies between -1 and 1. It does not establish causation.

10.1 Sensor calibration

The source voltage x and sensor output y in `data/calibration.csv` are entered as y ENTER x $\Sigma+$.

Keystrokes

Select model: f STAT CFIT MODL LINF.
Slope: f STAT CFIT SLOPE $\rightarrow m = 501.466667$.
Intercept: f STAT CFIT YINT $\rightarrow b = 0.777778$.
Correlation: f STAT CFIT CORR $\rightarrow r = 0.999997355$.

Therefore $\hat{y} = 501.466667x + 0.777778$ mV. At $x = 2.75$ V:

Keystrokes

2.75 f STAT CFIT FCSTY $\rightarrow$ 1379.8111 mV

To infer the input that corresponds to 1800 mV:

Keystrokes

1800 f STAT CFIT FCSTX $\rightarrow$ 3.587920 V

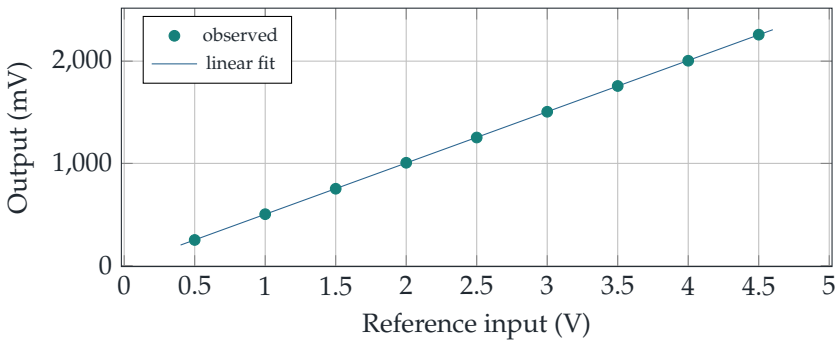


Figure 10.1: Calibration data and fitted line.

CHAPTER 11

Residuals, interpolation and extrapolation

A residual is $e_i = y_i - \hat{y}_i$. After fitting, compute each prediction with FCSTY and subtract it from the observed y. Plot or tabulate residuals against x.

Good residuals fluctuate around zero without curvature, funneling or runs. Curvature suggests a different model; widening spread suggests nonconstant uncertainty; a run may indicate drift. A very high $|r|$ does not guarantee clean residuals.

Interpolation predicts inside the observed x range; extrapolation predicts beyond it. The 2.75 V calibration is interpolation between 0.5 and 4.5 V. A 9 V prediction would be extrapolation and needs physical justification. Calibration equations are not licences to ignore sensor limits.

CHAPTER 12

Exponential regression: an RC discharge

For a discharging capacitor,

$$V(t) = V_0 e^{-t/\tau}.$$

The EXPF model has the form $y = be^{mx}$.

Enter `data/rc_discharge.csv`, then:

Keystrokes

f STAT CFIT MODL EXPF
f STAT CFIT SLOPE $\rightarrow m = -0.315976812$
f STAT CFIT YINT $\rightarrow b = 5.01423269$
f STAT CFIT CORR $\rightarrow r = -0.999978282$

Because $m = -1/\tau$, the fitted time constant is $\tau = 3.164789$ s. At 2.5 s, FCSTY gives 2.275815 V.

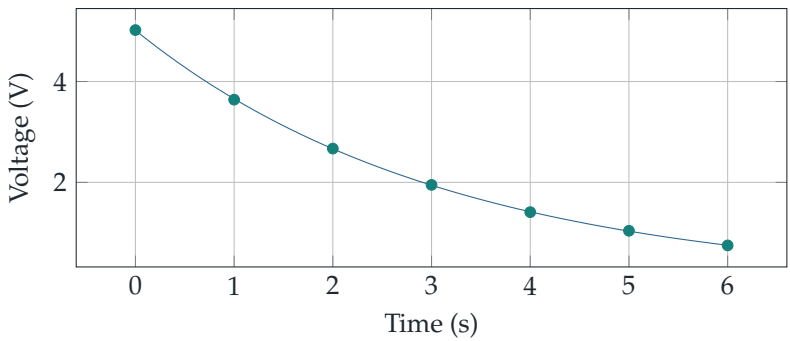


Figure 12.1: Exponential fit to RC discharge data.

CHAPTER 13

Power regression: the pendulum law

A simple pendulum predicts $T = 2\pi\sqrt{L/g}$, a power law $T = bL^m$ with $m = 1/2$. Fit `data/pendulum.csv` with PWRF:

Keystrokes

f STAT CFIT MODL PWRF
SLOPE $\to m = 0.499375064$ YINT $\to b = 2.00561250$ CORR $\to r = 0.999995397$

The exponent agrees closely with $1/2$. From $b = 2\pi/\sqrt{g}$, the implied $g = 4\pi^2/b^2 = 9.81444\ \mathrm{m/s^2}$.

CHAPTER 14

Logarithmic regression and model choice

The LOGF model is $y = m \ln x + b$ and requires $x > 0$. For measured sound level versus power in `data/sound_level.csv`, the fit is

$$\hat{y} = 4.32230522 \ln x + 79.7814413, \qquad r = 0.999941008.$$

This is physically sensible because decibel measures are logarithmic.

The BEST option tries candidate transformed models and selects the one with the largest absolute transformed correlation. It is a screening aid, not a scientific decision. With the narrow-range data in `data/best_trap.csv`, a linear fit has $r = 0.999288$, while an exponential fit has transformed $r = 0.999417$. BEST prefers exponential, but the distinction is tiny over the observed range. Mechanism, residuals and extrapolation behavior should decide.

Watch this

Transform restrictions are real. LOGF requires $x > 0$; EXPF requires $y > 0$; PWRF requires both $x > 0$ and $y > 0$. A domain error is evidence that the model is inapplicable or that the data need investigation—not an invitation to discard inconvenient points.

Part IV

Decision Workflows

CHAPTER 15

Expected cost and warranty planning

Expected value turns mutually exclusive outcomes into a long-run average:

$$E(C) = \sum_i p_i c_i.$$

For no claim (0.89, R0), minor repair (0.07, R120), major repair (0.03, R450) and replacement (0.01, R1200),

$$E(C) = 0.89(0) + 0.07(120) + 0.03(450) + 0.01(1200) = \text{R}33.90.$$

This is a portfolio budgeting figure, not the cost of a particular unit. Add administration, correlated failures and parameter uncertainty before using it as a price.

CHAPTER 16

Inventory under discrete demand

Demand $D = 0, 1, \ldots, 5$ has probabilities 0.05, 0.15, 0.30, 0.28, 0.15, 0.07. Unsold stock costs R80 per unit and shortage costs R220 per unit. For order quantity q,

$$C(q, D) = 80 \max(q - D, 0) + 220 \max(D - q, 0).$$

The expected costs for $q = 0$ through 5 are R558.80, R353.80, R193.80, R123.80, R137.80 and R196.80. The minimum is at $q = 3$.

The decision is transparent enough for a short worksheet: compute the cost for each demand in a row, multiply by its probability, and sum. The calculator handles the arithmetic; the decision model supplies the meaning.

CHAPTER 17

System reliability patterns

For independent components:

$$R_{\text{series}} = \prod_i R_i,$$
$$R_{\text{parallel}} = 1 - \prod_i (1 - R_i),$$
$$R_{\text{3-of-4}} = \binom{4}{3} p^3 (1-p) + p^4.$$

At $p = 0.9$, a 3-of-4 system has reliability 0.9477. Redundancy improves random independent reliability but may not protect against common-cause failure. State the independence assumption beside every result.

CHAPTER 18

Logistics: delivery-time prediction

The ten trips in `data/delivery.csv` produce

$$\widehat{\text{minutes}} = 1.45667643(\text{km}) + 11.5711595, \qquad r = 0.999308699.$$

At 20 km, FCSTY predicts 40.7047 minutes. The intercept represents fixed handling and urban overhead only within this operating context; it is not a universal travel-time constant. Weather, route class and load can create omitted-variable bias.

CHAPTER 19

Quality: tolerance is not standard deviation

For a roughly normal process with mean μ and standard deviation σ, the fraction between limits L and U is

$$P(L \le X \le U) = \Phi\left(\frac{U-\mu}{\sigma}\right) - \Phi\left(\frac{L-\mu}{\sigma}\right).$$

With $\mu = 20.001$ mm, $\sigma = 0.012$ mm and limits 19.97–20.03 mm, the model predicts 0.987276, or 98.7276%, in tolerance. The DM42 has no built-in normal CDF in the core statistics menu; evaluate the standardized-normal integral using the numerical integration workflow or a validated external table.

This is where the *DM42 Numerical Integration Handbook* takes over. Do not confuse an observed sample fraction with a fitted normal probability.

CHAPTER 20

A complete field workflow

Use the following sequence for any small engineering dataset:

1. Define the question, response, predictor, units and decision threshold.
2. Inspect the raw values and confirm decimal points and units.
3. Clear statistics with [f] [←] [CLΣ].
4. Enter one-variable data with [Σ+], or pairs as [y] [ENTER] [x] [Σ+].
5. Record n, mean and sample standard deviation before modelling.
6. Select a model from mechanism first, then examine r and residuals.
7. Predict only in a defensible range; label extrapolation.
8. Repeat a few calculations independently and preserve the raw data.
9. Report assumptions, units, rounding and the exact model form.

Part V

Reference

CHAPTER A

Keystroke atlas

Task	Exact route
Factorial	n f × N!
Combination	n ENTER r f × COMB
Permutation	n ENTER r f × PERM
Random number	f × RAN
Set seed	seed f × SEED
Clear statistics	f ← CLΣ
Add one value	x Σ+
Add a pair	y ENTER x Σ+
Remove an exact value/pair	re-enter it, then f Σ+ (Σ−)
Mean	f STAT MEAN
Sample deviation	f STAT SDEV
Sums	f STAT SUM
Weighted mean	f STAT WMEAN
Full sum registers	f STAT ↓ ALLΣ
Compact sum registers	f STAT ↓ LINΣ
Select fit	f STAT CFIT MODL, then model
Slope/intercept/r	f STAT CFIT, then SLOPE, YINT or CORR
Predict y from x	x f STAT CFIT FCSTY
Predict x from y	y f STAT CFIT FCSTX
Run named program	XEQ, select or spell label, ENTER
Insert program call	in program mode: XEQ, select or spell label, ENTER

CHAPTER B

What each result leaves in the stack

Function	X register	Y register
MEAN	$\bar{x}$	$\bar{y}$
SDEV	s_x	s_y
SUM	Σx	Σy
WMEAN	$\Sigma xy / \Sigma y$	prior stack result lifted normally
SLOPE	fitted m	prior stack result lifted normally
YINT	fitted b	prior stack result lifted normally
CORR	transformed-model r	prior stack result lifted normally

CHAPTER C

Model forms and domains

Model	Form	Domain/meaning
LINF	$y = mx + b$	additive change; any real x, y
LOGF	$y = m \ln x + b$	diminishing change; $x > 0$
EXPF	$y = be^{mx}$	constant percentage change; $y > 0$
PWRF	$y = bx^m$	scale law; $x > 0, y > 0$
BEST	largest $\lvert r \rvert$ after candidate transforms	screening only; compare residuals and mechanism

For EXPF and PWRF, YINT returns the coefficient b after back-transformation. SLOPE returns m. CORR is the correlation in the transformed coordinates used for the fit.

CHAPTER D

Common mistakes and recovery

Using [XEQ] before a menu function
Open the actual menu. [XEQ] is for named program/function execution and inserts XEQ in program mode.

Swapping paired data
Re-enter the pair and subtract it with $\Sigma-$, then add it as y [ENTER] x [$\Sigma+$].

Forgetting to clear old data
Use [f] [←] [CLΣ]; then re-enter the current dataset.

Reading the wrong stack level
MEAN, SDEV and SUM place the x result in X and the y result in Y.

Calling sample deviation population deviation
SDEV uses $n - 1$. Convert only if the entire finite population is truly observed.

Trusting BEST blindly
Compare residuals, domain restrictions and physical behavior.

Extrapolating without warning
State the data range and label predictions outside it.

Rounding intermediate probabilities
Retain the full internal value; round only the reported answer.

CHAPTER E

Formula sheet

$$
\begin{aligned}
{}_nP_r &= \frac{n!}{(n-r)!}, & \binom{n}{r} &= \frac{n!}{r!(n-r)!},\\
P(X=k)_{\text{bin}} &= \binom{n}{k} p^k (1-p)^{n-k}, & P(X=k)_{\text{Pois}} &= e^{-\lambda}\lambda^k/k!,\\
P(X=k)_{\text{hyp}} &= \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}, & \bar{x} &= \Sigma x/n,\\
s_x &= \sqrt{\frac{\Sigma x^2 - (\Sigma x)^2/n}{n-1}}, & \bar{x}_w &= \Sigma wx/\Sigma w,\\
\hat{y}_{\text{lin}} &= mx+b, & \hat{y}_{\text{exp}} &= be^{mx},\\
\hat{y}_{\text{pwr}} &= bx^m, & \hat{y}_{\text{log}} &= m\ln x + b.
\end{aligned}
$$

CHAPTER F

Dataset index

File	Purpose
`resistors.csv`	one-variable mean and sample deviation
`shafts.csv`	tolerance and outlier sensitivity
`calibration.csv`	linear regression and inverse prediction
`rc_discharge.csv`	exponential regression
`pendulum.csv`	power regression
`sound_level.csv`	logarithmic regression
`best_trap.csv`	warning against blind model selection
`delivery.csv`	operational linear regression
`inventory.csv`	expected-cost decision

CHAPTER G

Program entry and indirect storage note

Enter program mode with [f][R/S]. Each keystroke records a program instruction. To insert a subroutine call, press [XEQ]; that is why `XEQ` visibly appears on the line. This behavior is distinct from executing a shifted menu function during ordinary calculation.

For indirect storage in a program, the instruction is `STO IND 00`. On the calculator press [STO][·][IND][0][0]. The decimal-point key opens the indirect-parameter choice; [IND] selects it. Register 00 must contain the number of the destination register. For example, if R00 contains 17, `STO IND 00` stores X in R17. Validate the pointer range before a loop to avoid overwriting statistical registers.

CHAPTER H

Glossary

Bias
systematic difference between an estimator's average and the target.

Correlation
standardized linear association; not proof of causation.

Expected value
probability-weighted long-run average.

Extrapolation
prediction outside the observed predictor range.

Independent events
events for which learning one does not alter the probability of the other.

Interpolation
prediction inside the observed predictor range.

Residual
observed response minus fitted response.

Sample standard deviation
spread estimate with denominator $n - 1$.

Seed
starting state for a pseudorandom sequence.

Weighted mean
mean in which observations carry unequal weights.

CHAPTER I

Validation and provenance

All printed numeric examples are checked in double precision by the supplied validation script:

```
validation/validate.py
```

Each assertion uses a tolerance tighter than the displayed rounding. The script covers probability examples, descriptive summaries, all four regression forms, prediction, reliability, warranty cost, inventory choice and normal-model yield.

Calculator behavior and menus were checked against the current SwissMicros DM42 quick reference and user manual, and against the current Free42 command implementation. In particular, the source confirms the $n - 1$ standard deviation, X/Y result order, weighted-mean calculation, transformed regression definitions, BEST selection rule, and random seed handling. See the bibliography and `SOURCES.md`.

CHAPTER J

Probability pattern selector

Clue	Model	First calculation
Ordered choices	permutation	${}_nP_r$
Unordered choices	combination	$\binom{n}{r}$
Fixed n, independent, two outcomes	binomial	$\binom{n}{k}p^k(1-p)^{n-k}$
Without replacement from finite lot	hypergeometric	divide favorable selections by $\binom{N}{n}$
Counts in time/space at stable rate	Poisson	$e^{-\lambda}\lambda^k/k!$
At least one	complement	$1-P(0)$
Independent series	product	$\prod R_i$
Independent parallel	failure complement	$1-\prod(1-R_i)$

CHAPTER K

Core summation-register map

The compact LINΣ layout stores the following six accumulators beginning at R11. Treat them as owned by the statistics system while a dataset is active.

Register	Accumulator	Register	Accumulator
R11	Σx	R14	Σy^2
R12	Σx^2	R15	Σxy
R13	Σy	R16	n

ALLΣ extends the occupied region through R23: $\Sigma \ln x$, $\Sigma (\ln x)^2$, $\Sigma \ln y$, $\Sigma (\ln y)^2$, $\Sigma (\ln x \ln y)$, $\Sigma (x \ln y)$ and $\Sigma (y \ln x)$, in that order. Do not mix application storage with R11–R23 unless you deliberately relocate the statistics base register.

CHAPTER L

Reporting template

Copy this compact structure into a lab book or service report:

1. **Question:** what quantity or decision is required?
2. **Data:** source, date, units, sample size and exclusions.
3. **Method:** model and calculator route; state transformations.
4. **Summary:** mean, sample deviation and range where relevant.
5. **Fit:** equation, correlation, residual pattern and data range.
6. **Result:** value with units and sensible significant figures.
7. **Assumptions:** independence, constant rate, normality or mechanism.
8. **Decision:** action threshold and consequence of being wrong.

Never report a regression as only "$r = \ldots$". Include the model form, coefficients, units, range and residual assessment.

CHAPTER M

Reusable program listings

The short `DIE` program appears in Chapter 5. The longer `MCPI` program accepts the trial count N in X and returns $4m/N$. It deliberately uses numbered registers below the default statistics block.

Listing M.1: MCPI: unit-square Monte Carlo estimate of π

```
LBL "MCPI"
STO 00
0
STO 01
STO 02
LBL 00
RAN
X^2
RAN
X^2
+
1
X<>Y
X<=Y?
GTO 01
GTO 02
LBL 01
1
STO+ 01
LBL 02
1
STO+ 02
RCL 02
RCL 00
X>Y?
GTO 00
4
RCL 01
x
RCL 00
/
RTN
```

Enter program mode with f R/S. For menu instructions such as RAN, enter the same physical route used in run mode: f × RAN. Use XEQ only when the

listing itself contains an `XEQ` call. Exit program mode with f R/S; run with XEQ, select `MCPI`, ENTER.

The returned estimate varies with the seed. Before a reproducible run, enter a nonzero seed and use f × SEED. Verify the program first with $N = 10$ and confirm the result lies between 0 and 4; then use larger N.

CHAPTER N

Calculator verification checklist

After a reset, firmware change or migration between DM42 and Free42, use these quick checks:

1. 8 ENTER 3 f × COMB must give 56.
2. Clear sums; add 1, 2, 3; MEAN must put 2 in X and SDEV must put 1 in X.
3. Clear sums; enter $(x, y) = (1, 3), (2, 5), (3, 7)$ as y ENTER x Σ+; LINF must give slope 2, intercept 1 and correlation 1.
4. Seed with the same nonzero value twice; the first subsequent RAN value must repeat.
5. In program mode, press XEQ and select a label; the line must show an `XEQ` instruction.
6. Enter 17 in R00, place a test value in X, and execute `STO IND 00`; R17 must contain the test value.

These tests check conventions and state. They are not a substitute for reviewing firmware release notes.

CHAPTER O

Verified answer index

Example	Verified result
$P(X \geq 2)$, $n = 20$, $p = .025$	0.0882417
At least 9 successes of 10, $p = .98$	0.9838224
Hypergeometric at least one	0.3530395
Poisson $P(X \geq 4)$, $\lambda = 2.4$	0.2212771
Resistor mean; sample s	10.0050; 0.0206706
Calibration m, b, r	501.466667; 0.777778; 0.999997355
RC m, b, τ	−0.315976812; 5.01423269; 3.16478920
Pendulum m, b, g	0.499375064; 2.00561250; 9.81444351
Sound-level LOGF m, b	4.32230522; 79.7814413
Series; parallel; 3-of-4 reliability	0.945847; 0.9936; 0.9477
Expected warranty cost	R33.90
Minimum inventory cost	$q = 3$, R123.80
Normal-model in-tolerance fraction	0.9872758

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Colophon

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